Welcome to CSCI 585 Database Systems



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Mon 2-4 pm

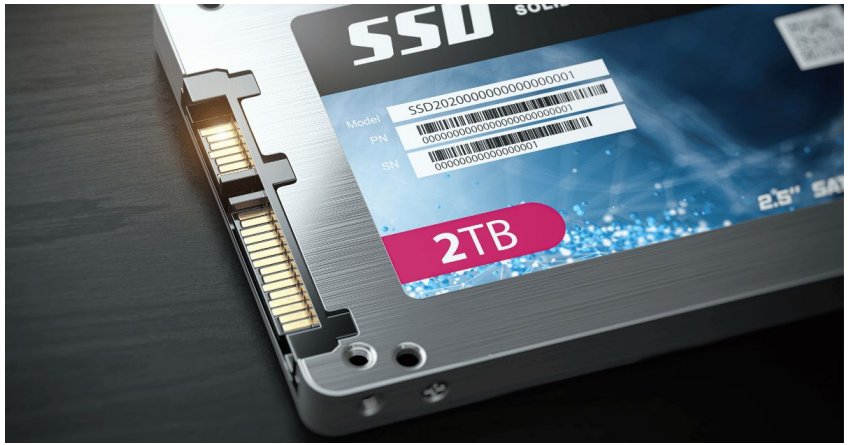
Database and DBMS

- Database is an integrated collection of data, usually stored on secondary storage, typically describing the activities of one or more related organizations.

- A database management system (DBMS) is a collection of software/programs designed to assist in maintaining and utilizing large collections of data.

Secondary Storage: Disk and Solid State Disk





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Before Computers

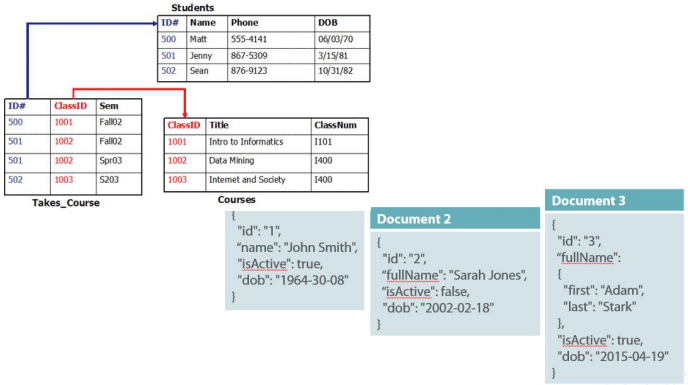


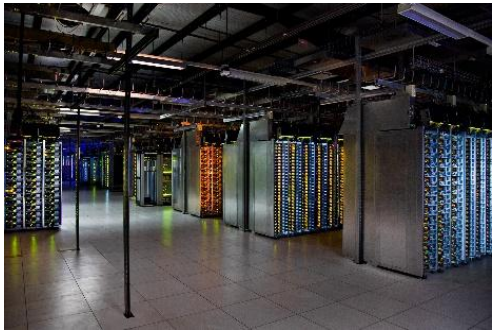
Database



DBMS

Today's Systems







Demanding applications, E.g., Facebook

- High Volume: 2.4 billion user profiles, 250 billion friend connections, 3.13 trillion likes, 37 billion tagged locations, 240 billion photos, ….
- High Variety: Mix of data types: Structured records, multimedia content, text.

Database



DBMS

Future Biological Computers

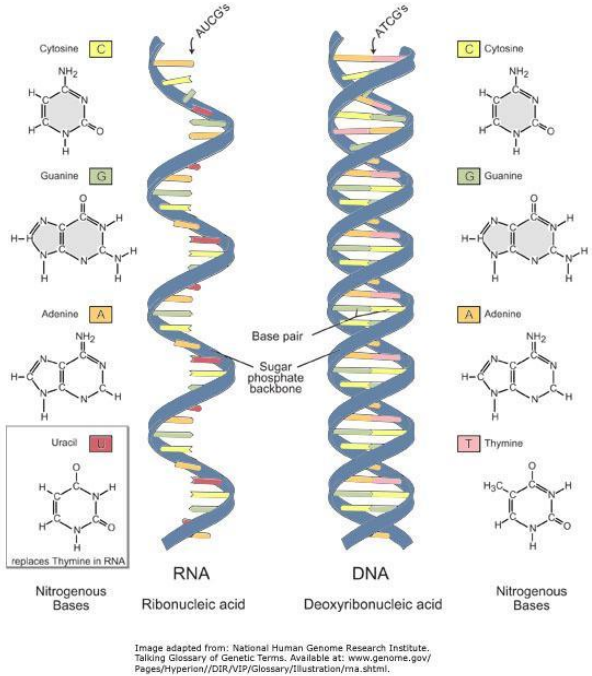
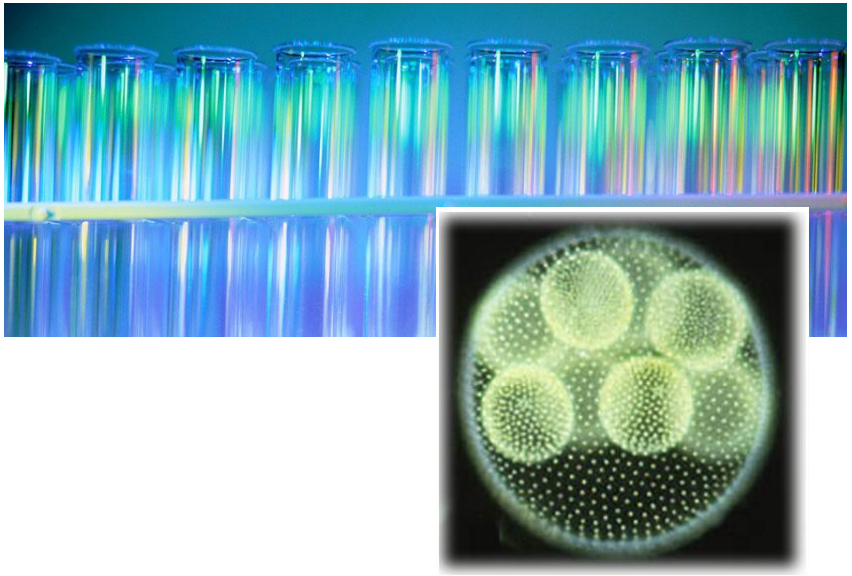


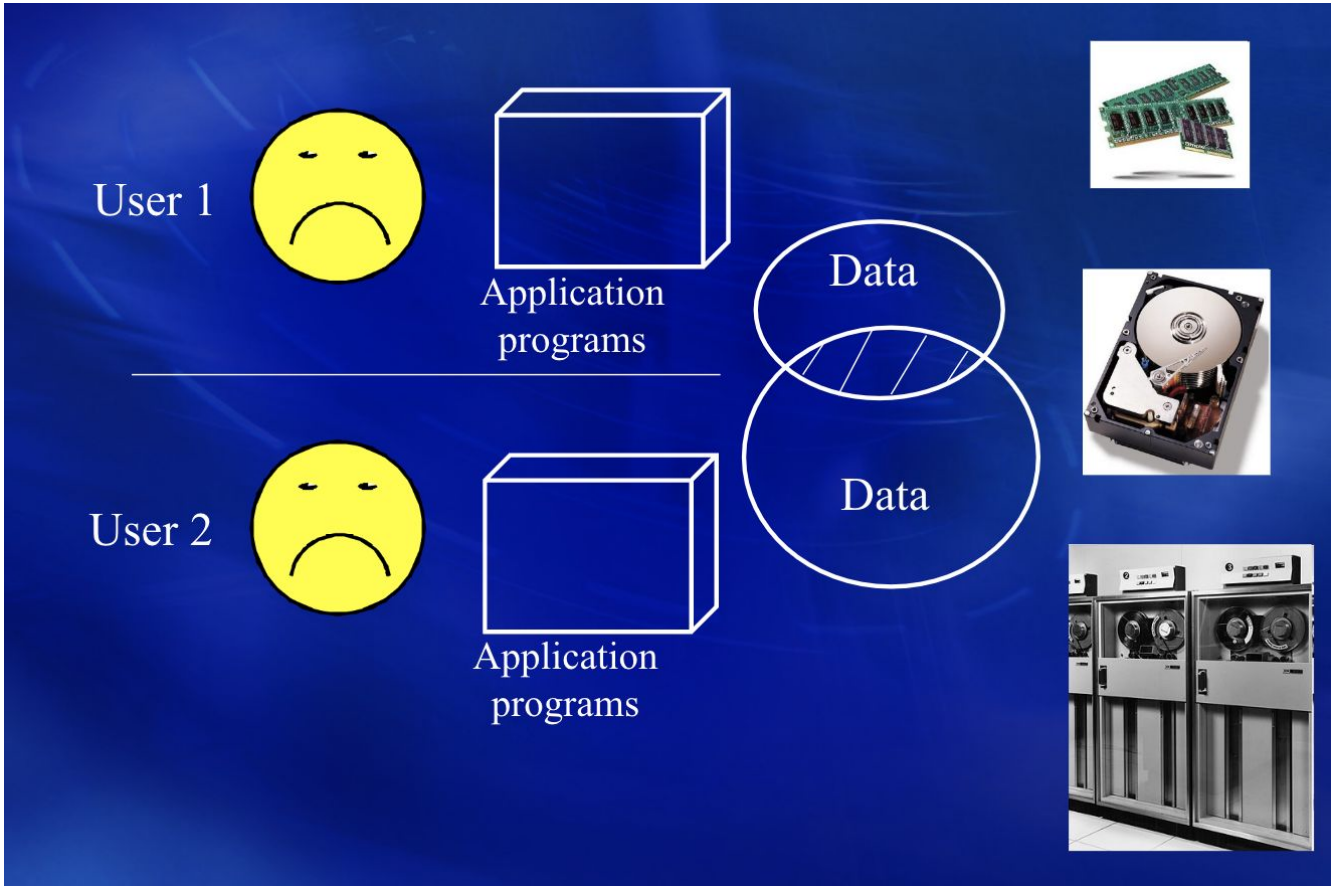
Image adapted from: National Human Genome Research Institute. Talking Glossary of Genetic Terms. Available at: www.genome.gov/Pages/Hyperion//DIR/VIP/Glossary/Illustration/rna.shtml.

Database

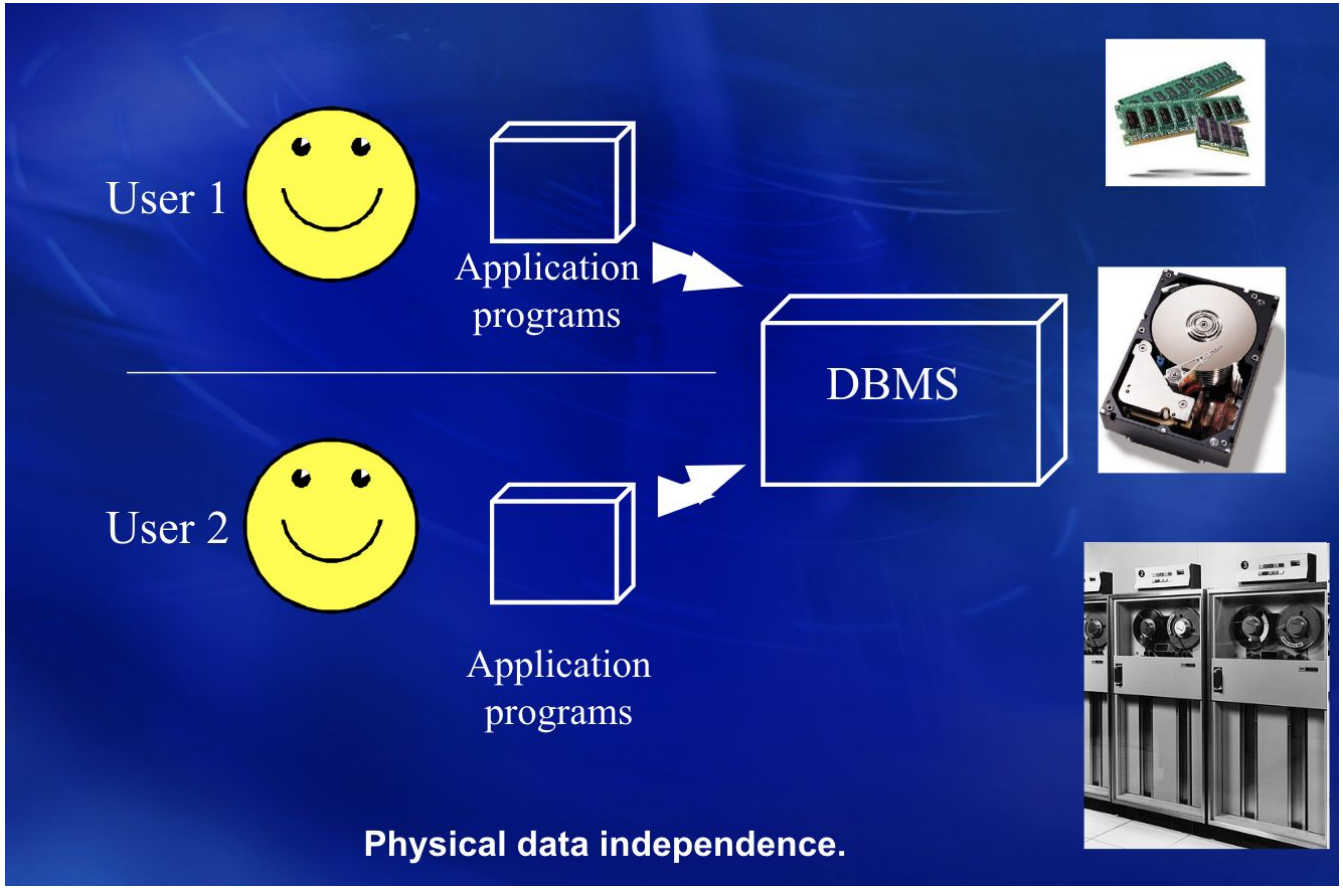


DBMS

Before DBMS


User 1
Application programs
Data
User 2
Application programs
Data

After DBMS


User 1
Application programs
DBMS
User 2
Application programs
Physical data independence.

Why A DBMS?

- Reduced application development time
- Data independence: Application programs not dependent on data representation and storage details
- Data sharing: data is better utilized (discovered and reused), redundancy of data is minimized
- Data integrity and consistency: one may enforce consistency constraints on data, e.g., number of seats sold ≤ number of seats on the plane × 1.1
- Centralized control: DBA tunes the database to balance user's needs
- Security: mechanisms to prevent unauthorized access. These mechanisms are based on content instead of file-oriented approach.
- Concurrency control: avoids undesirable race conditions that arise with simultaneous access/updates to data
- Crash recovery: ensures the integrity of data in the presence of failures

Data Models



Build a database to keep the medical records of U.S. residents



Conceptual

Logical

Physical





Modern DBMSs: Guest Lecturers

- eBay NuGraph, Dr. Hieu Nguyen, September 15



- Databricks, Dr. Haoyu Huang.



CSCI 585: Research Manuscripts

B.S./B.A

Read Textbooks

Broad Knowledge



M.S.

Read Manuscripts

Specialized Knowledge

Ph.D.

Write Manuscripts

Create Knowledge

Grading

Assessment Tool (assignments)	Points	% of Grade
Exam 1, Oct 12, 2026	100	50%
Exam 2, Nov 30, 2026	100	50%
TOTAL		**100%**

Exams 1 and 2 cover material from the assigned papers **only**.

TAs & Office Hours



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QUESTIONS